

Expired by Climate — 4-minute video script

Target length 3:55 · 332 spoken words · record slides from the paper's Figures 1–4

Time	On screen	Narration (about 130 words a minute)
0:00–0:20	Close-up of a medicine strip: "Store below 30 °C".	"This line is printed on medicines all over the hot half of the world. It isn't just advice. It's a climate forecast — and it was made decades ago. We checked whether it still holds."
0:20–0:55	Photo or drawing of a stability chamber; the three conditions 25/60, 30/65, 30/75.	"Before a medicine is approved, batches sit in a chamber for a year or more: 25 degrees for temperate markets, 30 for hot ones. Those numbers came from weather records, turned into one figure — the mean kinetic temperature, the effective temperature a medicine experiences — plus a safety margin."
0:55–1:25	Timeline: 1971 Haynes → 1998 Grimm → 2006 WHO Zone IVb → 2024 Japan check → Nov 2026 ICH Q1.	"The climate has warmed since. The only recent re-check, for Japan, found no problem. As far as we found, nobody had re-checked hot countries — or hot cities inside temperate ones — decade by decade. And the global stability guideline is being finalised this year."
1:25–2:20	Figure 1 (conceptual model), then the three equations briefly.	"So we re-ran the regulators' own method on 45 years of data: ERA5 reanalysis for 10 cities and NOAA weather stations for 8. For each year we compute the effective temperature and humidity, subtract them from the test condition, and track that safety margin. We fixed our hypotheses before downloading anything, checked our code against the textbook example, checked Japan, and measured our own method's bias against hourly data."
2:20–3:10	Figure 2 (dumbbell chart): every dot moves right; highlight Houston crossing the line. Then Figure 3 (humidity).	"Every one of the 18 places warmed — by 0.3 to 1.3 degrees. 7 now sit above the temperature their medicines are tested at, including Houston and Tampa, inside the United States, which crossed the 25-degree line between the 1990s and today. In humid coastal cities like Chennai and Colombo, the air is already wetter than the 30-degree, 65-percent chamber. And Tokyo, our control, stays safely inside — just as the Japanese study found."
3:10–3:40	Figure 4 (shelf-life curve) and three icons: regulators, surveillance, open tool.	"Each degree over the line costs a typical medicine about a tenth of its shelf life. This doesn't prove any product failed — buildings are often cooled. It shows where the standard and the climate have drifted apart, so regulators know where to look first. Our tool and data are open."
3:40–3:55	Back to the strip: "Store below 30 °C".	"A standard is a forecast. Forecasts need checking."

Recording tips: speak the hook over the strip close-up; keep each figure on screen while you describe it; end before 4:00 — the limit is strict.